HUARAVITO LTEMS

Long-Term Ecological Monitoring Site

Date: February 27, 2026

Location: (10.104583, -84.564639)
Piedades Sur, San Ramón, Costa Rica

Data Annex Integrated

PART I · Data Inventory Overview

Dataset ID	Dataset Name	Years	Core Variables	Frequency	Format
D01	Microclimate Time Series	2022–2026	Temp (°C), Humidity (%), Time	Hourly / Transect	CSV
D02	Forest Structure Census	2023–2026	DBH, Height, Species ID	Annual	CSV
D03	Soil & Organic Layer Monitoring	2023–2026	Organic Depth (cm), Moisture	Biannual	CSV
D04	Biodiversity Observations	2022–2026	Species, Taxon, Date, Location	Continuous	CSV / iNaturalist

PART II · Dataset Specifications

D01 — Microclimate Time Series

A. File Description

Continuous temperature and humidity records collected via fixed-station sensors and linear transect validation protocol.

B. Variable Definitions

Variable	Unit	Description
Temp	°C	Ambient air temperature at 1.5 m height
Humidity	%	Relative humidity
Timestamp	ISO 8601	Date-time record

C. Data Integrity Notes

Cross-validated with analog mercury thermometer. Sensor drift checks quarterly.

D02 — Forest Structure Census

A. File Description

Annual tree census within core and buffer zones.

B. Variable Definitions

Variable	Unit	Description
DBH	cm	Diameter at breast height (1.6 m)
Height	m	Estimated tree height
Species_ID	Text	Taxonomic identification

C. Data Integrity Notes

Standardized DBH tape measurement. Re-measured by two observers.

D03 — Soil & Organic Layer Monitoring

A. File Description

Organic layer depth and soil moisture measurements across permanent plots.

B. Variable Definitions

Variable	**Unit**	**Description**
Organic_Depth	cm	Leaf litter + humus depth
Soil_Moisture	%	Volumetric soil moisture content

C. Data Integrity Notes

Measured using soil probe. Averaged across 3 subsamples per station.

D04 — Biodiversity Observations

A. File Description

Species occurrence records integrated with iNaturalist project database.

B. Variable Definitions

Variable	Unit	Description
Species	Text	Scientific name
Taxonomic_Group	Text	Family or higher classification
Observation_Date	Date	Date of record
Latitude/Longitude	Decimal Degrees	Georeferenced coordinate

C. Data Integrity Notes

Community-verified observations. Research-grade filter applied.

PART III · Analytical Indicators

1. Biogenic Cooling Index (BCI) = [(T_matrix − T_forest) / T_matrix] × 100

2. Tree Density per Hectare = (Number of Trees / Plot Area m^2) × 10,000

3. Mean DBH = Σ(DBH) / n

4. Species Accumulation Rate = dS/dt

5. Organic Layer Change = Depth_year(n) − Depth_year(n−1)

Indicators operationalize the hypothesis of **Autopoietic Resilience** by linking microclimate buffering, structural complexity, and biodiversity persistence.